

-

Grok

Invalid Date

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- ω_n
- $\zeta = 1$
- $\omega_n + \zeta = 1$ ζ

1.

Bode

- ω_n -20dB/dec
- $\zeta = 1$

- $\Rightarrow \zeta \geq 1$
- $\zeta = 1$ ω_n

2.

- $\zeta = 0.65 \sim 0.8$ $\zeta \approx 0.707$
 – $\zeta = 0.707$ Butterworth 4%
 – $\zeta = 1$
- ω_n
 – $\omega_n \approx 6 \sim 12$ rad/s 1~2 Hz
 – $\frac{\omega_n}{\omega_n}$
 – $\frac{\omega_n}{\omega_n}$

- -
- c
- ζ
- $\omega_n +$ +
- **PID** / /

- ITAE
- H_2 / H_∞
-

MATLAB `fmincon` ga `simulink` Python `scipy.optimize`

3.

```
“python # Python    import numpy as np from scipy import signal import
matplotlib.pyplot as plt
wn = 8.0 # rad/s    zeta = 0.707 #
sys = signal.TransferFunction([wn**2], [1, 2zeta*wn, wn**2])
```

Bode